

Lec9

Media Queries

Media Queries are CSS techniques used to apply different styles based on the device's screen size, resolution, orientation, or other characteristics.

Basic Syntax

```
@media (condition) {  
  /* CSS rules */  
}
```

Why Use Media Queries?

A website should look good on:

- Mobile Phones
- Tablets
- Laptops
- Desktop Screens

Media queries allow the layout to change automatically for different devices.

Example

```
@media (max-width: 768px) {  
  body {  
    background-color: lightblue;  
  }  
}
```

If the screen width is **768px or less**, the background becomes light blue.

Types of Media Queries

max-width

Applies styles when the screen width is **less than or equal to** a value.

```
@media (max-width: 768px) {  
  h1 {  
    color: red;  
  }  
}
```

min-width

Applies styles when the screen width is **greater than or equal to** a value.

```
@media (min-width: 768px) {  
  h1 {  
    color: green;  
  }  
}
```

Range Query

```
@media (min-width: 600px) and (max-width: 900px) {  
  body {  
    background: yellow;  
  }  
}
```

Common Breakpoints

Device	Width
Mobile	up to 480px
Tablet	481px – 768px
Small Laptop	769px – 1024px
Desktop	1025px and above

Example

```
/* Mobile */
@media (max-width:480px){
  body{
    font-size:14px;
  }
}
/* Tablet */
@media (min-width:481px) and (max-width:768px){
  body{
    font-size:16px;
  }
}
/* Desktop */
@media (min-width:769px){
  body{
    font-size:18px;
  }
}
```


CSS Transitions

A **transition** creates a smooth change between two states of an element.

Without transition:

```
button:hover{  
    background:red;  
}
```

With transition:

```
button{  
    background:blue;  
    transition: background 1s;  
}
```

```
button:hover{  
    background:red;  
}
```

Transition Properties

transition-property

Specifies which property will change.

transition-property: background-color;

transition-duration

Specifies how long the transition takes.

transition-duration: 2s;

transition-delay

Waits before starting.

transition-delay: 1s;

transition-timing-function

Controls speed pattern.

transition-timing-function: ease;

Types:

- linear
- ease
- ease-in
- ease-out
- ease-in-out

Diagram:

linear : -----

ease-in : ----->

ease-out : <-----

ease-in-out : <----->

CSS Transforms

Transforms change an element's **size, position, rotation, or shape**.

Transform Properties

Translate

Moves an element.

```
transform: translate(100px,50px);
```

Diagram:

Before After

[A] ---> [A]

Rotate

Rotates an element.

```
transform: rotate(45deg);
```

Diagram:

Before After

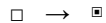


Scale

Changes size.

```
transform: scale(1.5);
```

Diagram:

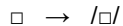


Skew

Tilts an element.

```
transform: skew(20deg);
```

Diagram:



Multiple Transforms

```
transform: translateX(50px) rotate(45deg);
```

Hover Example

```
.box{  
transition:0.5s;  
}
```

```
.box:hover{  
transform:scale(1.2);  
}
```

CSS 3D Transforms

3D transforms create depth effects.

Properties

rotateX()

Rotates around X-axis.

```
transform: rotateX(60deg);
```

rotateY()

Rotates around Y-axis.

```
transform: rotateY(60deg);
```

rotateZ()

Same as normal rotation.

```
transform: rotateZ(45deg);
```

translateZ()

Moves element closer or farther.

```
transform: translateZ(100px);
```

Perspective

Adds depth.

```
.container{  
  perspective:1000px;  
}
```

Without perspective:

Flat rotation

With perspective:

Looks like real 3D rotation

3D Card Flip Example

```
div class="card">
```

```
  Front
```

```
</div>
```

```
.card{
```

```
  transition: 1s;
```

```
}
```

```
.card:hover{
```

```
  transform: rotateY(180deg);
```

```
}
```

CSS Animators

Animations allow elements to move automatically without user interaction.

They use:

1. `@keyframes`
2. `animation` property

Basic Animation

```
@keyframes move{  
  from{  
    transform:translateX(0);  
  }  
  to{  
    transform:translateX(300px);  
  }  
}  
.box{  
  animation: move 2s;  
}
```

Keyframes with Multiple Steps

```
@keyframes colorChange{  
  0%{  
    background:red;  
  }  
  50%{  
    background:yellow;  
  }  
  100%{  
    background:green;  
  }  
}
```

Animation Properties

animation-name

animation-name: move;

animation-duration

animation-duration: 3s;

animation-delay

animation-delay: 2s;

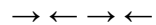
animation-iteration-count

animation-iteration-count: infinite;

animation-direction

animation-direction: alternate;

Diagram:



animation-timing-function

animation-timing-function: ease-in-out;

Shorthand

animation: move 3s ease infinite alternate;

Task

Convert a desktop page into a responsive page and create animated buttons, cards, and hero section in page